CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination

November 2012

CL209 Structural Analysis-I

Date: 06.11.2012, Tuesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Discuss advantages and disadvantages of indeterminate structures. [07]

Q - 2 (a) Prove in case of a thin shell, Circumferential stress, $f_c = Pd/2t$ and Longitudinal stress, $f_l = Pd/4t$. [04]

(b) Attempt any Two [10]

(i) A short hollow pier of 1.2 m square section outside and 1.0 m square inside is subjected to a direct load of 120 kN along its outer edge point. Determine the final stress at the base of pier. Draw neat sketch of stress distribution diagram.

(ii) Find the maximum and minimum stress intensities in a short C.I. Column of hollow circular section attached with a projection of bracket carrying a load of 60 kN. The external and internal diameter of the column are 300 mm and 250 mm respectively. The load line is off the axis of the column by the 300 mm.

(iii) Find the maximum and minimum stress intensities induced on the base of a masonary wall 6 meter high, 4 m wide and 1.5 m thick subjected to a horizontal wind pressure of 1500 N/mm^2 acting on 4 meter side. The density of masonary may be taken as 22400 N/mm^3.

Q - 3 (a) Derive the expression for change in volume of thin cylindrical shell due to the internal pressure. [04]

(b) Attempt any Two [10]

(i) A mild steel tube is used as a strut, having 25 mm external diameter, 2.5 mm thickness and 3 m length. The strut is hinged at both ends. Calculate the collapsing load by Euler's formula. E = 2 x 105 N/mm^2.

(ii) A rectangle column section, size b x D is fixed at both ends. Determine the limiting length of the column, so that critical stress is 7 N/mm^2. Take E = 20000 N/mm^2.

(iii) A Tee-section, 150 mm wide x 75 mm deep, thickness of flange 9 mm, thickness of web 8.4 mm, is used as a strut, 3 m long, ends hinged. Calculate safe axial load by Rankine's formula, using a factor of safety of 3. Rankine's constants are fc= 315 N/mm^2, a = 1/7500.

SECTION – II

Q - 4 (a) A three hinged parabolic arch hinged at the supports and at crown has a span of 24 m and a central rise of 4 m. It carries a concentrated load of 50 kN at 18 m from left support and an UDL of 30 kN/m over the left half portion. Determine the moment, [07] thrust and radial shear at a section 6 m from the left support.

Q - 5 (a) A three hinged arch of span 'l' and rise 'h' caqrries a uniformly distributed load of 'w' pre unit run over the whole span, Show that the horizontal thrust at each support is $wl^2/8h$. [04]

(b) **Attempt any Two** [10]

(i) Two wheel loads of 16kN and 8kN at a fixed distance 2m apart cross a beam of 10m span. Draw the influence line for B.M. and S.F. for a point 4m from the left abutment. Also find maximum B.M and S.F. at that point.

(ii) Draw I.L.D. for shear force and bending moment at a section 3m from left end of a simply supported beam 12m long.
Also find maxi. S.F. and B.M. at this section due to u.d.l. of 2kN/m and 5m length roll from left to right.

(iii) Two wheel loads, 50 kN each spaced at 4m centre to centre, cross a girder of 8m span. Find absolute maximum B.M. in the beam.

Q - 6 **Attempt any Two** [14]

(i) Using double integration method, obtain slopes at supports and deflection at the centre for a simply supported beam of length 4 m subjected to a central point load of 240 kN. Take c/s of beam = 100 mm x 200 mm and E = 0.7 x 105 N/mm^2.

(ii) A beam 'AB' of length 5 m is loaded with UDL of 20 kN / m for a span of 2 m starting at 1 m distance from left support. Find out slopes at both supports and deflection at starting and end point of UDL.

(iii) A beam 'AB' of 8 m span is simply supported at the ends. It is loaded with a point load of 50 kN at a distance 1 m from end 'A' and a UDL of 10 kN / m of span 3 m from the end 'B'. Determine the deflection at starting of UDL, Maximum Deflection and slope at 'A'. Take $E = 2*10^5\ N/mm^2$ and $I = 10{,}500\ cm^4$.

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